Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 (Generated practice paper - Set 11)

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours**

### Materials

For this paper you may use:
- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| 11 | |
| TOTAL | |

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**Answer all questions in the spaces provided.**

**0 1** A student measured the mass of carbon stored per hectare in three land uses in the same region.

**Table 1**

| Land use | Carbon in vegetation / t $ha^{-1}$ | Carbon in soil dead organic matter / t $ha^{-1}$ |
|---|---|---|
| Mature woodland | 120 | 95 |
| Permanent pasture | 3 | 80 |
| Arable (ploughed annually) | 2 | 45 |

**0 1 . 1** Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

**[1 mark]**

**Answer** ☐ t/ha

**0 1 . 2** Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

**[2 marks]**

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**0 1 . 3** Suggest **two** ways the arable farmer could increase soil carbon storage.

**[2 marks]**

**1** .....

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**2** .....

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**5**

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**0 2 . 1** Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

**[1 mark]**

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

**0 2 . 2** Explain why rainwater collection is increasingly important in cities such as those in India.

**[2 marks]**

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**0 2 . 3** Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

**[2 marks]**

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**0 3 . 1** Complete Table 1 to show one method of controlling each source of railway noise.

**[4 marks]**

**Table 1**

| Source of railway noise | Control method | |
|---|---|---|
| Wheel vibration on the track | | |
| Pantograph turbulence | | |
| Wheel squeal on corners | | |
| Braking squeal | | |

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**0 3 . 2** Explain why road traffic noise affects more people than any other form of noise pollution.

**[1 mark]**

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**0 3 . 3** Describe **three** ways of reducing road traffic noise reaching residents.

**[3 marks]**

**1** .....

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**2** .....

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**3** .....

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**0 3 . 4** Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

**[2 marks]**

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**10**

**0 4** Table 1 shows the annual emissions of two pollutants in the UK in 1970 and 2015.

**Table 1**

| Pollutant | Emissions in 1970 / thousand tonnes | Emissions in 2015 / thousand tonnes | Main source in 2015 |
|---|---|---|---|
| Sulfur dioxide | 6 400 | 220 | Energy industries |
| Nitrogen oxides | 2 700 | 930 | Road transport |

**0 4 . 1** Calculate the percentage reduction in sulfur dioxide emissions between 1970 and 2015.

**[2 marks]**

**Answer** ..... %

**0 4 . 2** Suggest **three** reasons why sulfur dioxide emissions fell by a much larger percentage than nitrogen oxide emissions.

**[3 marks]**

**1** .....

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**2** .....

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**0 4 . 3** Explain why a reduction in NOx emissions would also reduce concentrations of tropospheric ozone and PANs.

**[2 marks]**

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**0 4 . 4** Explain why lichens are used to monitor sulfur dioxide pollution.

**[3 marks]**

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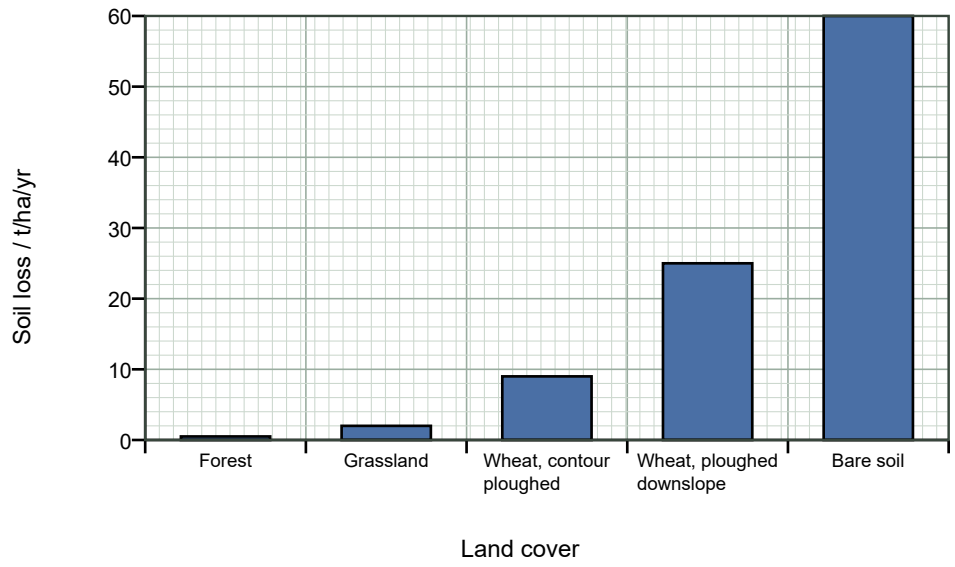
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**0 5** Figure 1 shows the annual soil loss measured on experimental plots with different vegetation cover on a 10% slope in a region with heavy seasonal rain.

**Figure 1**




**0 5 . 1** Describe the pattern shown in Figure 1.

**[2 marks]**

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**0 5 . 2** Explain **four** ways in which vegetation reduces the rate of soil erosion.

**[4 marks]**

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**0 5 . 3** Explain why ploughing up and down a slope causes more erosion than contour ploughing.

**[2 marks]**

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**0 5 . 4** Explain why the region's climate makes the soil vulnerable to **both** wind and water erosion.

**[2 marks]**

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**0 6** Two insecticides, X and Y, were tested on a pest insect and on a predatory ground beetle that feeds on the pest. Table 1 shows the results.

**Table 1**

| Insecticide | $LD_{50}$ for pest / mg $kg^{-1}$ | $LD_{50}$ for ground beetle / mg $kg^{-1}$ | Half-life in soil / days | Solubility in water |
|---|---|---|---|---|
| X | 5 | 6 | 180 | Low |
| Y | 8 | 150 | 4 | High |

**0 6 . 1** Explain what is meant by $LD_{50}$.

**[1 mark]**

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**0 6 . 2** Using Table 1, explain which insecticide is more likely to cause indirect effects on the pest population through the food chain.

**[3 marks]**

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**0 6 . 3** Suggest **two** environmental problems that could be caused by insecticide Y that are less likely with insecticide X.

**[2 marks]**

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**0 6 . 4** Explain why the half-life of insecticide X makes it more likely to bioaccumulate.

**[2 marks]**

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**0 6 . 5** Give **two** reasons why a systemic insecticide may be preferred to a contact insecticide.

**[2 marks]**

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**0 7 . 1** Which of the following statements about stratospheric ozone is correct? Tick one box.

**[1 mark]**

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

**0 7 . 2** Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

**[2 marks]**

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**0 7 . 3** Explain why HCFCs cause less ozone depletion than CFCs.

**[2 marks]**

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**0 8** Table 1 shows the energy used per passenger-kilometre by different forms of transport when carrying a typical load.

**Table 1**

| Mode of transport | Typical occupancy | Energy use / MJ per passenger-km |
|---|---|---|
| Car (petrol, 1 occupant) | 1 | 2.9 |
| Car (petrol, 4 occupants) | 4 | 0.7 |
| Bus (diesel) | 25 | 0.6 |
| Electric train | 200 | 0.3 |
| Bicycle | 1 | 0.06 |
| Short-haul aircraft | 120 | 2.4 |

**0 8 . 1** Calculate the percentage reduction in energy per passenger-km achieved by carrying four people in a car instead of one.

**[1 mark]**

**Answer** ☐ %

**0 8 . 2** Explain why bulk transport (buses, trains) uses less energy per passenger-km than cars.

**[2 marks]**

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**0 8 . 3** Suggest **three** reasons why people continue to use cars despite the energy advantages of public transport.

**[3 marks]**

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**0 8 . 4** Explain how integrated transport systems and Active Traffic Management reduce energy use.

**[4 marks]**

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**0 9** A mining company is assessing a copper deposit beneath farmland.

**0 9 . 1** Describe what each of the following survey techniques detects.

**[3 marks]**

**Gravimetry** .....

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**Magnetometry** .....

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**Seismic surveys** .....

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**0 9 . 2** Explain why trial drilling is normally carried out only after the remote sensing surveys.

**[2 marks]**

**0 9 . 3** Explain how a rise in the world price of copper can turn part of the **resource base** into a **reserve**.

**[2 marks]**

**0 9 . 4** Explain **four** environmental impacts of extracting the copper by opencast mining.

**[4 marks]**

**0 9 . 5** Describe how the site could be restored after closure, and explain why restoration is planned before mining begins.

**[4 marks]**

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**15**

**1 0** Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

**Table 1**

| **Location** | **Latitude** | **Mean solar energy in June / kWh $m^{-2}$ $day^{-1}$** | **Mean solar energy in December / kWh $m^{-2}$ $day^{-1}$** |
|---|---|---|---|
| Sahara desert | 25°N | 7.8 | 4.2 |
| Southern England | 51°N | 5.0 | 0.7 |
| Northern Norway | 69°N | 4.6 | 0.0 |

**1 0 . 1** Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

**[1 mark]**

**Answer** ..... times

**1 0 . 2** Explain why northern Norway receives more solar energy in June than in December.

**[2 marks]**

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**1 0 . 3** Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

**[2 marks]**

**1 0 . 4** Explain why the low energy density of solar power is a problem for large-scale generation.

**[1 mark]**

**1 0 . 5** Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

**[9 marks]**

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**Write an essay on one of the following topics.**

**1 1 . 1** Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

**[25 marks]**

**OR**

**1 1 . 2** Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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**END OF QUESTIONS**